

Q17. Add a UNIQUE constraint on the conference name.

alter table "ResearchDB".Conferences add constraint unique_c_name unique (conference_name);

```
59  
60 alter table "ResearchDB".Conferences  
61 add constraint unique_c_name unique (conference_name);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 125 msec.

Q19. Create a schema named Academic .

create schema Academic;

```
65  
66 create schema Academic;
```

Data Output Messages Notifications

CREATE SCHEMA

Query returned successfully in 68 msec.

Q20. Create a table named Projects with appropriate SQL data types (SERIAL, VARCHAR, DATE, and NUMERIC).

create table "ResearchDB".Projects (
p_id serial primary key,
p_name varchar(20) not null,
p_date date,
budget numeric(10,2)
);

```
67  
68 create table "ResearchDB".Projects (  
69 p_id serial primary key,  
70 p_name varchar(20) not null,  
71 p_date date,  
72 budget numeric(10,2)  
73 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 69 msec.

Q21. Create an index on the last_name column of the Authors table.

```
create index last_name_idx on "ResearchDB".Authors(last_name);
```

```
74
75 create index last_name_idx
76 on "ResearchDB".Authors(last_name);
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 68 msec.

Q22. Create an index on the publication_year column of the Papers table

```
create index p_y_index on "ResearchDB".Papers(publication_year);
```

```
77
78 create index p_y_index
79 on "ResearchDB".Papers(publication_year);
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 116 msec.

Q23. Create a new user named research_student.

```
create user research_student
with password '1234';
```

```
81 create user research_student
82 with password '1234';
83
```

Data Output Messages Notifications

CREATE ROLE

Query returned successfully in 61 msec.

Q24. Grant SELECT permission on the Papers table to the new user.

```
grant select
on "ResearchDB".Papers
to research_student;
```

```
84 grant select
85 on "ResearchDB".Papers
86 to research_student;
87
```

Data Output Messages Notifications

GRANT

Query returned successfully in 65 msec.

Q25. Revoke the SELECT permission from the user.

```
revoke select  
on "ResearchDB".Papers  
from research_student;
```

```
87  
88 revoke select  
89 on "ResearchDB".Papers  
90 from research_student;
```

Data Output Messages Notifications

REVOKE

Query returned successfully in 98 msec.